BS2591 MACROECONOMIC THEORY AND APPLICATIONS

Tutorial 2
The 3-equation model

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29 October 2026 · Lecture 3 problem set · Carlin and Soskice, Chapter 3

WARM-UP · THE MODEL IN ONE SLIDE

Three equations, one central bank

IS · DEMAND

The lever. Today's rate moves output **next** period: $y_t = A - a\,r_{t-1}$.

$$
\begin{aligned}
y &= \frac{1}{1 - c_1(1-t)}\left[c_0 + (a_0 - a_1 r) + G\right] && (1.10)\\
&= k\left[c_0 + (a_0 - a_1 r) + G\right] && (1.11)\\
&= k(c_0 + a_0 + G) - k a_1 r && (1.12)\\
&= A - ar, && \text{(IS curve)}
\end{aligned}
$$

where $A \equiv k(c_0 + a_0 + G)$ and $a \equiv k a_1$; k is the multiplier.

Chapter 1 p. 30

PC · SUPPLY

The constraint. Inflation only falls if output is below y_e.

$$
\begin{aligned}
\pi_t &= \pi_t^{E} + \alpha(y_t - y_e)\\
\pi_t &= \pi_{t-1} + \alpha(y_t - y_e). && \text{(adaptive expectations Phillips curve)}
\end{aligned}
$$

Chapter 2 p. 97

MR · POLICY

The best response: minimise the loss subject to the PC.

$$y_t - y_e = -\alpha\beta(\pi_t - \pi^{T})$$

Appendix 3.5.1

The recipe for every shock: forecast next period's PC → find where it meets the MR → read the interest rate off the IS.

Carlin and Soskice (2024). Equation numbers and pages as in the book; summary in Box 3.1.

LECTURE 3 · QUESTION 1

Explain what is meant by the central bank's loss function. How are the central bank's preferences reflected in the loss function?

$$L = (y_t - y_e)^2 + \beta(\pi_t - \pi^T)^2$$

What it measures

The cost of being away from equilibrium output and from the inflation target.

Why squared

Above and below target are equally bad, and big misses hurt more than small ones.

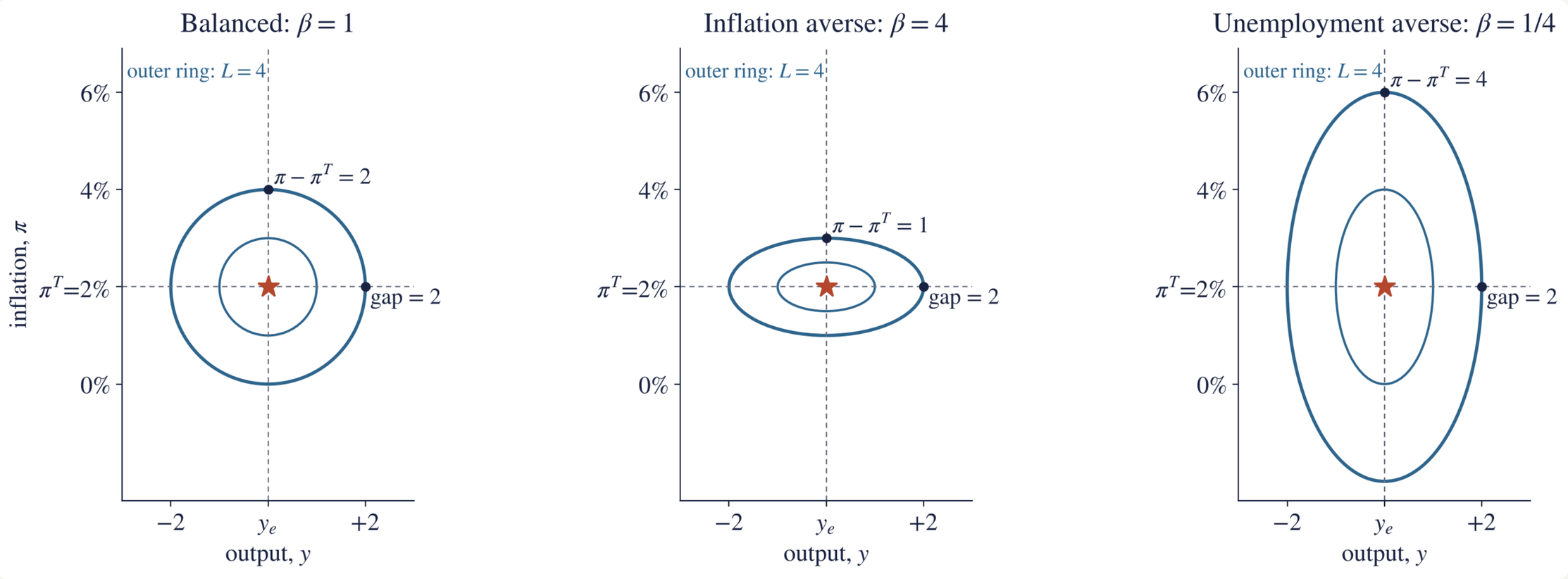
Where preferences sit

β is the weight on inflation. β > 1: inflation averse. β < 1: unemployment averse.

Textbook pp. 129–131. Example with $\pi^T = 2\%$: inflation of 4% and of 0% give the same loss, $\beta \times 2^2 = 4\beta$.

QUESTION 1 · READING B OFF THE SHAPE

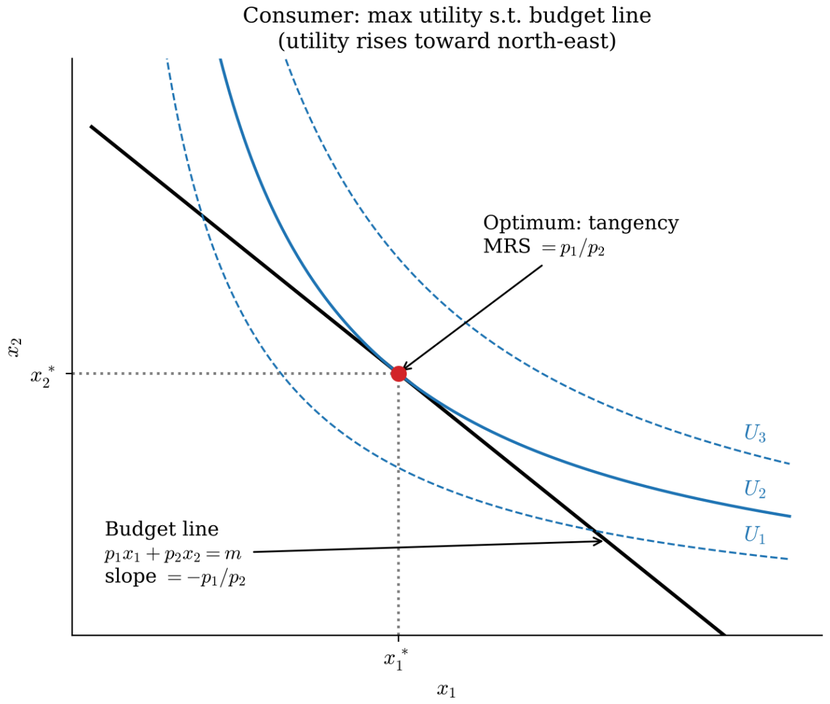
Same loss, different shapes: circles and ellipses

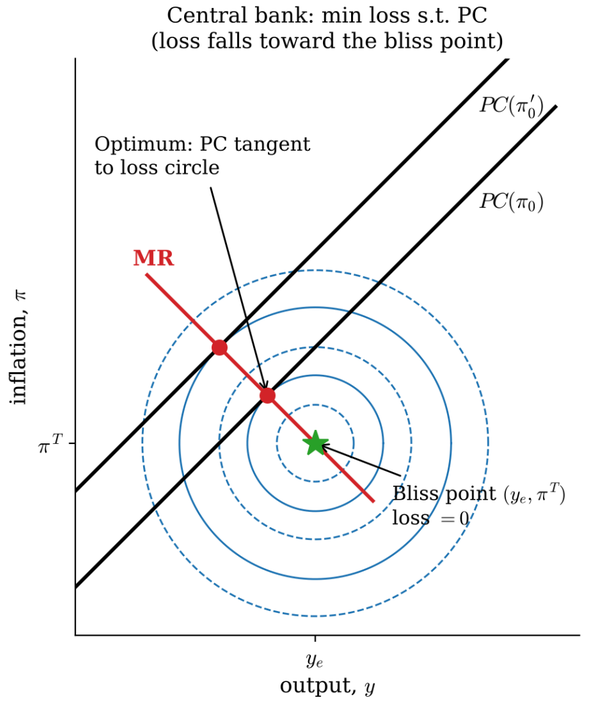


Each outer ring is $L = 4$. Compare Fig. 3.2 in the answers and Fig. 3.5 in the textbook.

FROM LOSS TO POLICY · AN ANALOGY YOU ALREADY KNOW

The Phillips curve is the central bank's budget line





Consumer: best bundle is where the budget line touches the highest indifference curve.

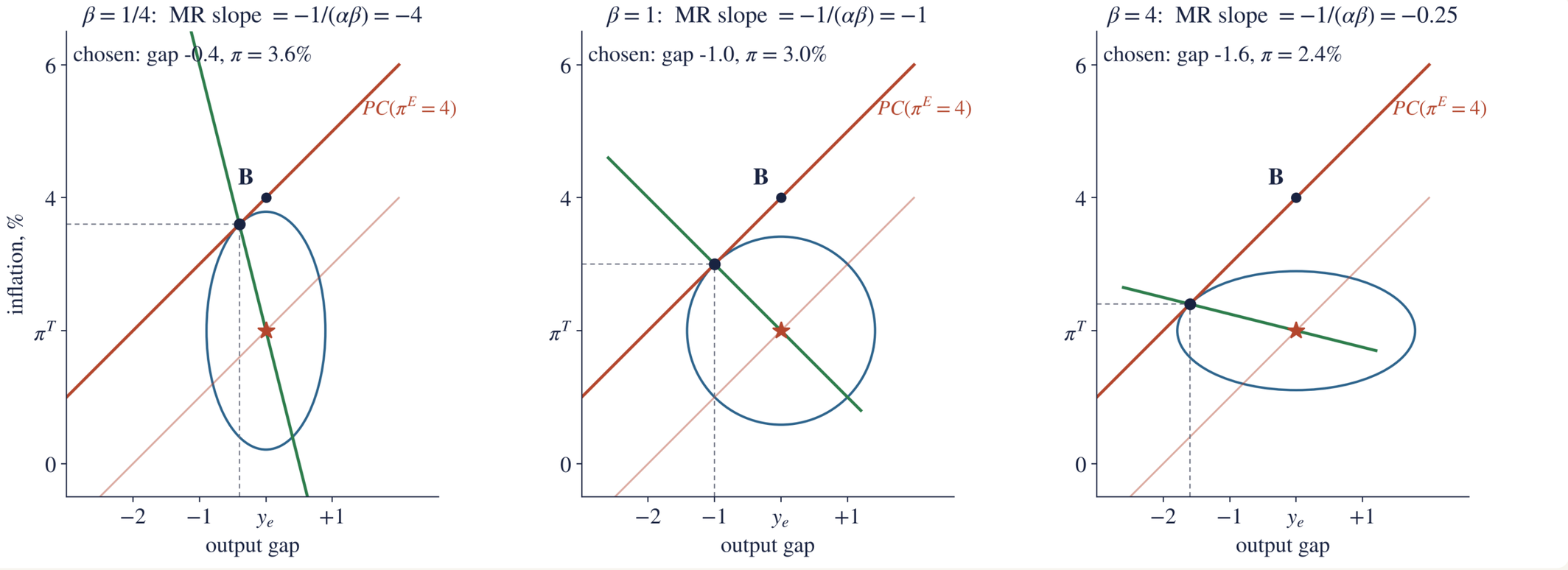
Central bank: best point is where the PC touches the smallest loss circle.

Join the tangency points for every PC: that line is the **MR**.

Textbook: Figs. 3.6 and 3.7. The bank cannot pick any point it likes: this period's expectations fix which PC it is on.

LECTURE 3 · QUESTION 2

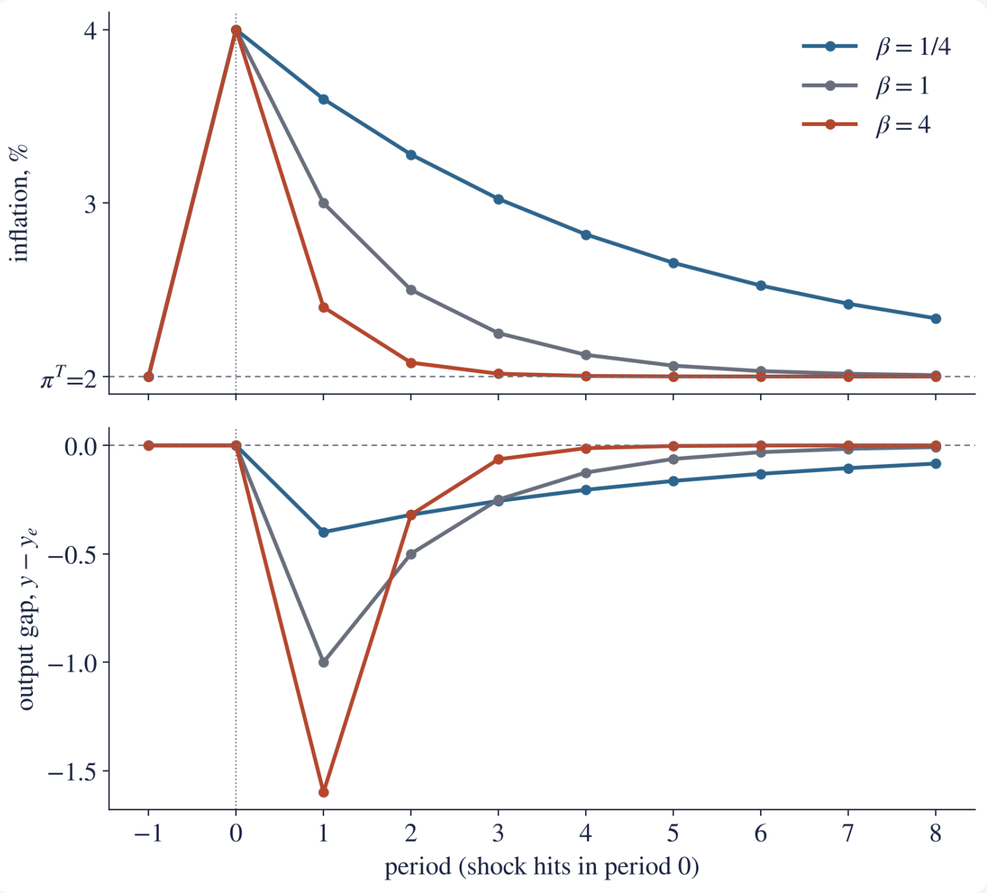
In which of the three cases (β < 1, β = 1, β > 1) will the central bank reduce inflation back to target quickest after an inflation shock? Is there any downside to adopting this policy stance?



Same shock in every panel: inflation jumps to 4% at B, α = 1. Only β differs.

QUESTION 2 · PERIOD BY PERIOD

Higher β: a deeper recession, a faster return



β	$y_1 - y_e$	π_1	periods until π within 0.25 of target
1/4	−0.4	3.6%	10
1	−1.0	3.0%	3
4	−1.6	2.4%	2

Each period the inflation gap shrinks by the factor $1/(1 + \alpha^2\beta)$: 0.8 for β = 1/4, 0.5 for β = 1, 0.2 for β = 4.

Answer: β > 1. The inflation-averse bank gets back to target quickest.

Simulated with α = 1, shock π_0 = 4%, π^T = 2%. Output falls one period after the shock because of the lag in the IS.

QUESTION 2 · IS THERE A DOWNSIDE?

Speed has a price: a sharper, though shorter, recession

Deeper first hit

With β = 4 output falls by 1.6 at once, four times the fall with β = 1/4. More firms lay off workers.

But shorter

The gap closes in a couple of periods. With β = 1/4 a smaller gap lasts much longer.

Same end point

Every bank ends at (y_e, π^T) with $r = r_s$. β changes the path, never the destination.

So which is better?

Unclear. A deep slump can scar workers' skills and lower y_e itself (hysteresis, Chapter 4). The model leaves this out.

LECTURE 3 · QUESTION 3

Draw the 3-equation model and give a detailed period by period description of the adjustment process for the case where the economy is hit by a permanent negative aggregate demand shock.

Five steps that work for every shock:

1

Which curve moves? For one period or for good?

2

Forecast next period's PC: $\pi^{E}_{t+1} = \pi_t$.

3

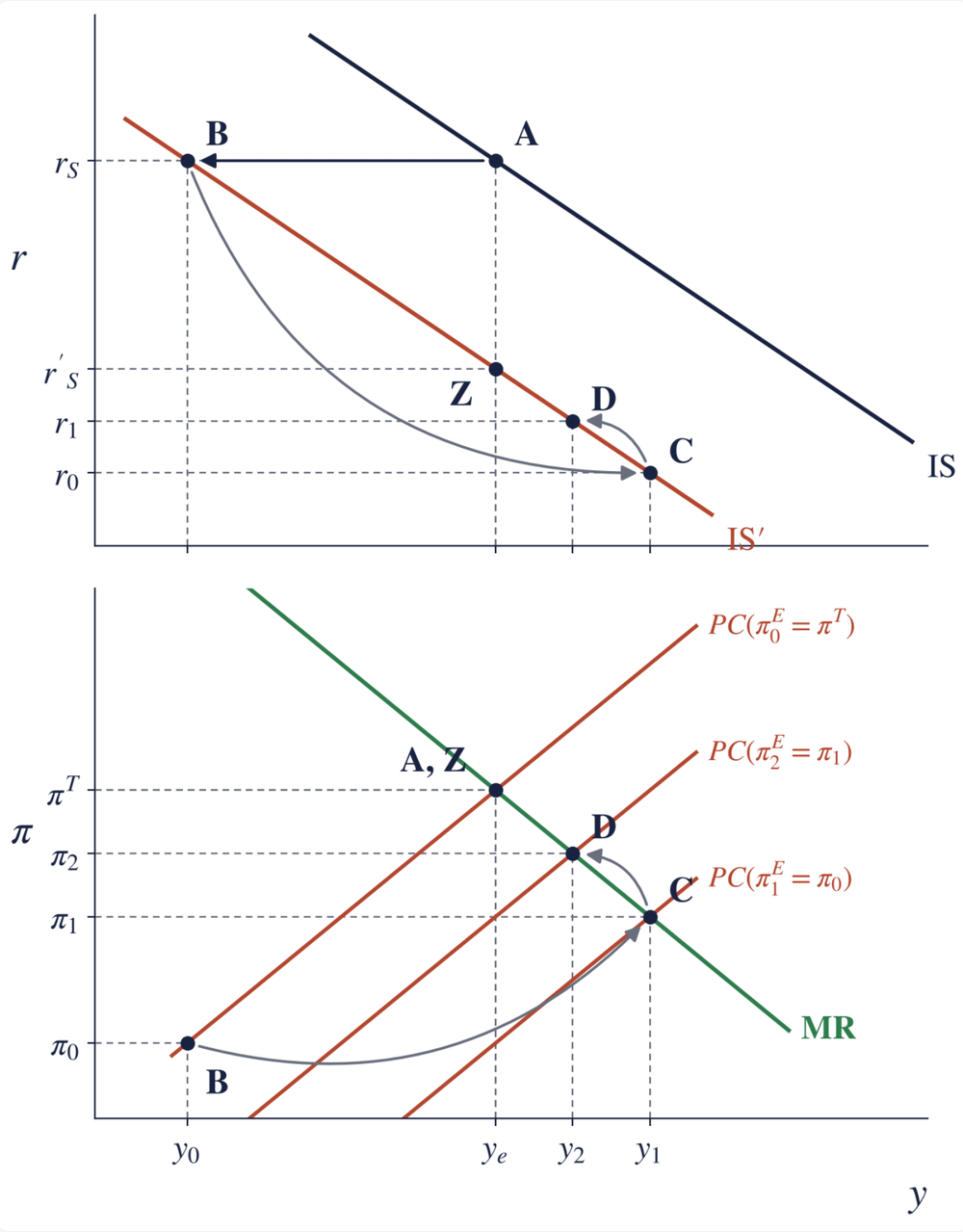
Where that PC meets MR gives the output target.

4

Read *r* off the IS that will hold next period.

5

Next period: arrive, update π, repeat until Z.



QUESTION 3 · PERMANENT NEGATIVE DEMAND SHOCK

From A to Z in three moves

Period 0 · A → B

IS shifts left for good: output falls to y_0, inflation to $\pi_0 < \pi^T$. The bank forecasts PC($\pi^E_1 = \pi_0$), aims for C on MR and **cuts** r to r_0 on the new IS′.

Period 1 · B → C

The cut now works: output rises to y_1 **above** y_e, so inflation rises to π_1. The bank forecasts PC($\pi^E_2 = \pi_1$), aims for D and **raises** r to r_1.

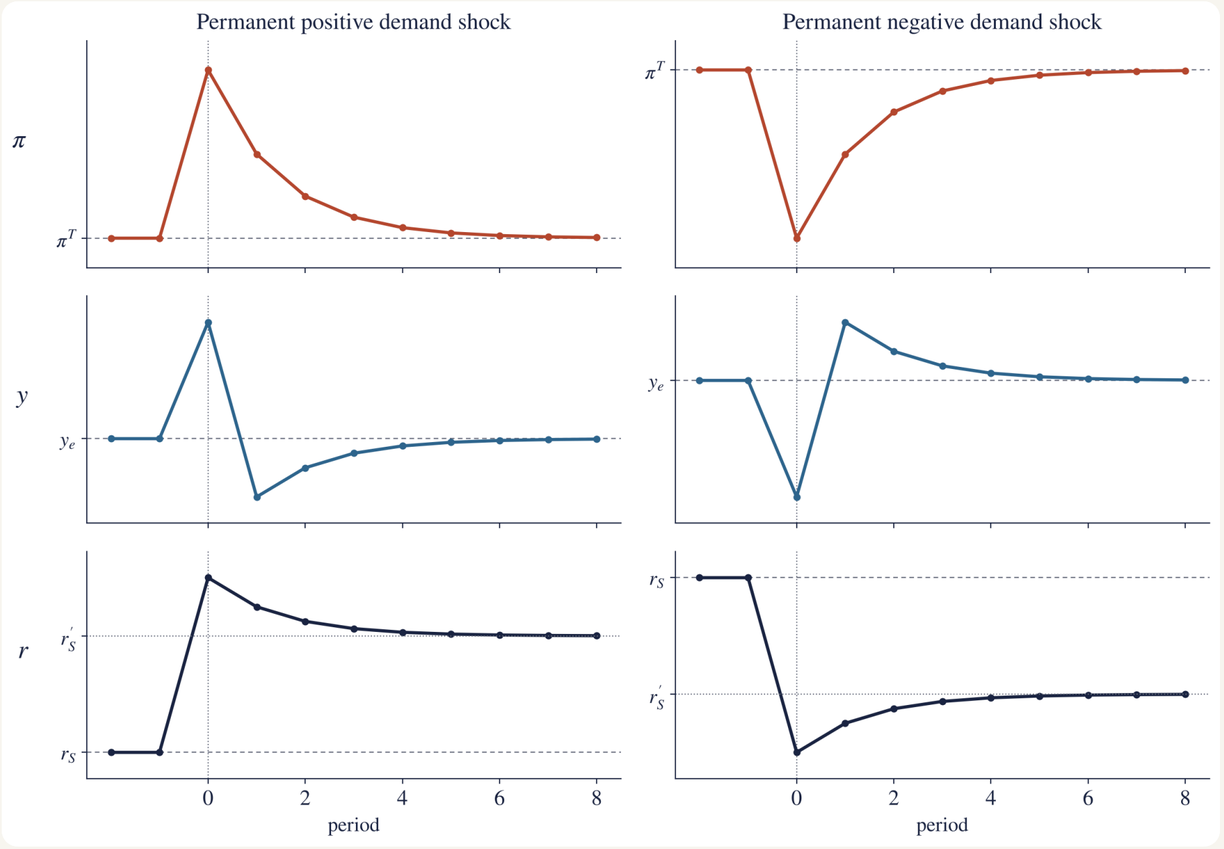
Period 2 onwards · C → D → … → Z

Rates rise step by step along IS′. At Z: y_e, π^T and a new, **lower** stabilising rate r'_s.

Why a boom at C? On MR, $\pi < \pi^T$ goes with $y > y_e$: to lift inflation back up, the bank needs a positive gap.

LECTURE 3 · QUESTION 4

Draw the impulse response functions for output, inflation and the real interest rate after a permanent negative and a permanent positive aggregate demand shock.



π

Jumps on impact, then returns slowly to π^T.

y

Jumps, then **overshoots** the other way: the bank needs the opposite gap to bring π back.

r

Moves by more than the shift in r_s, then settles at a **new** stabilising rate: higher after a positive shock, lower after a negative one.

The negative shock is a mirror image.

Simulated with α = β = *a* = 1. The shock hits in period 0; the rate set in period 0 affects output from period 1.

LECTURE 3 · QUESTION 5

Redo questions 3 and 4 for the cases of all temporary and permanent, negative and positive, demand and supply shocks.

Shock (2 × 2 × 2)	What moves	Bank's first move	Final y_e	Final r_s	Worked through in
AD temporary +	IS right, one period	Raise r (old IS)	same	same	CS Fig. 3.10
AD temporary –	IS left, one period	Cut r (old IS)	same	same	mirror of 3.10
AD permanent +	IS right for good	Raise r (new IS)	same	higher	CS 3.4, 3.11; Q4
AD permanent –	IS left for good	Cut r (new IS)	same	lower	Q3, Q4; ZLB: 3.13
AS temporary +	PC down, one period	Cut r	same	same	mirror of 3.9
AS temporary –	PC up, one period	Raise r	same	same	CS Fig. 3.9; Q2
AS permanent +	y_e up: MR, PCs right	Cut r	higher	lower	CS Fig. 3.14
AS permanent –	y_e down: MR, PCs left	Raise r	lower	higher	mirror of 3.14

LECTURE 3 · QUESTION 6

In the view of many economists, the Great Recession started off a large negative permanent aggregate demand shock. However, contrary to the predictions of the standard 3-equation model, many economies, including the UK, did not recover to the original equilibrium.

(a) Use the 3-equation model diagrams to show how the economy can fall into a deflation trap.

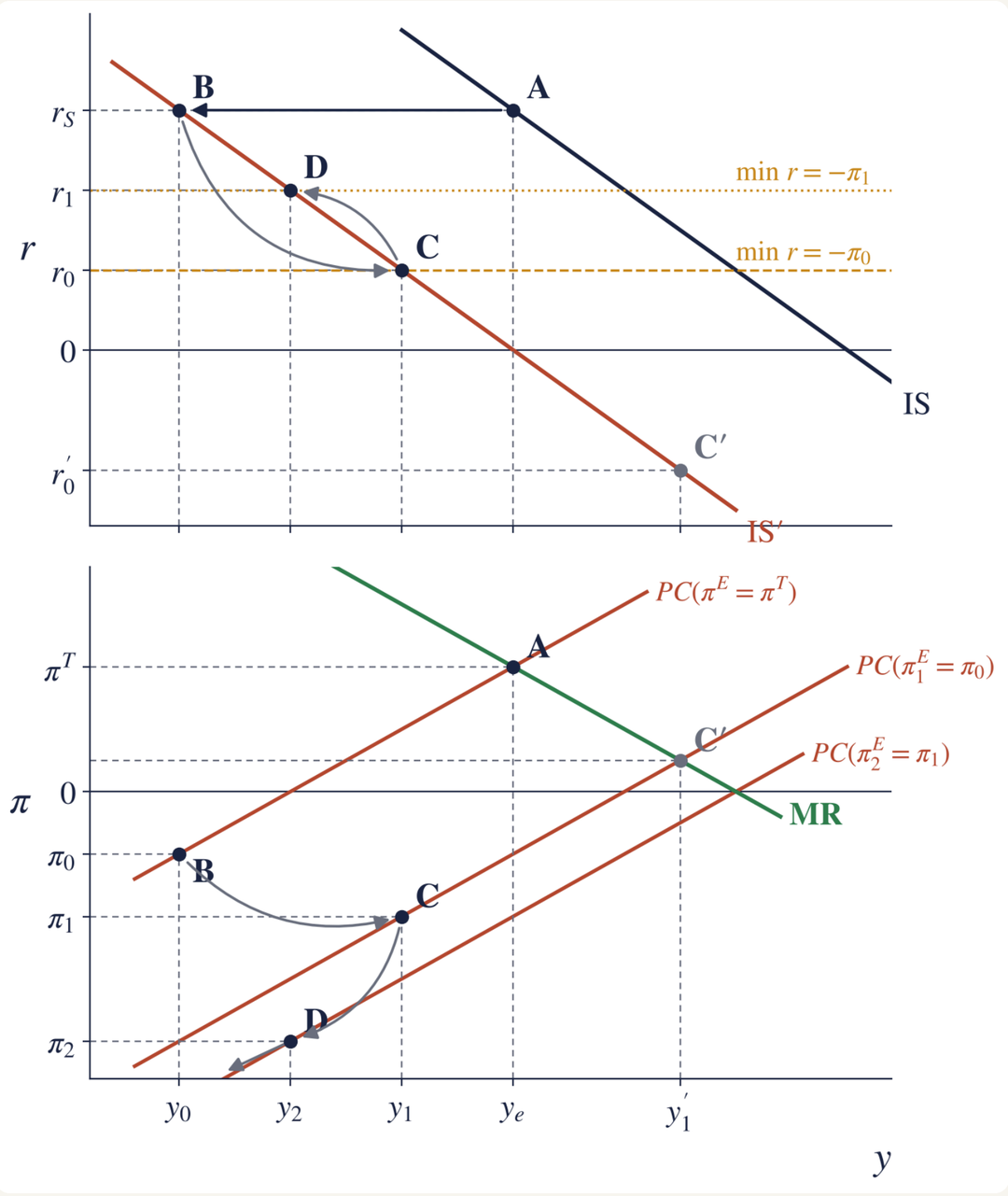
(b) Explain, with reference to the diagram, how the central bank or government can intervene to escape the trap.

(c) In the Great Recession, what policy mistakes may have kept the UK from returning to the original equilibrium?

$i = r + \pi^E$ and $i \geq 0 \Rightarrow \min r = -\pi^E$

The key fact: once inflation turns negative, the lowest real rate the bank can reach goes **up**.

Textbook: section 3.3.4 and Figs. 3.12–3.13. Question wording in (c) shortened.



QUESTION 6(A) · THE DEFLATION TRAP

When cutting rates is not enough

Period 0 · A → B

A big shift of IS to IS′. Output collapses and inflation turns negative: $\pi_0 < 0$. To reach C′ on MR the bank needs $r'_0 < 0$, but the floor is $-\pi_0$. It can only set $r_0 = -\pi_0$.

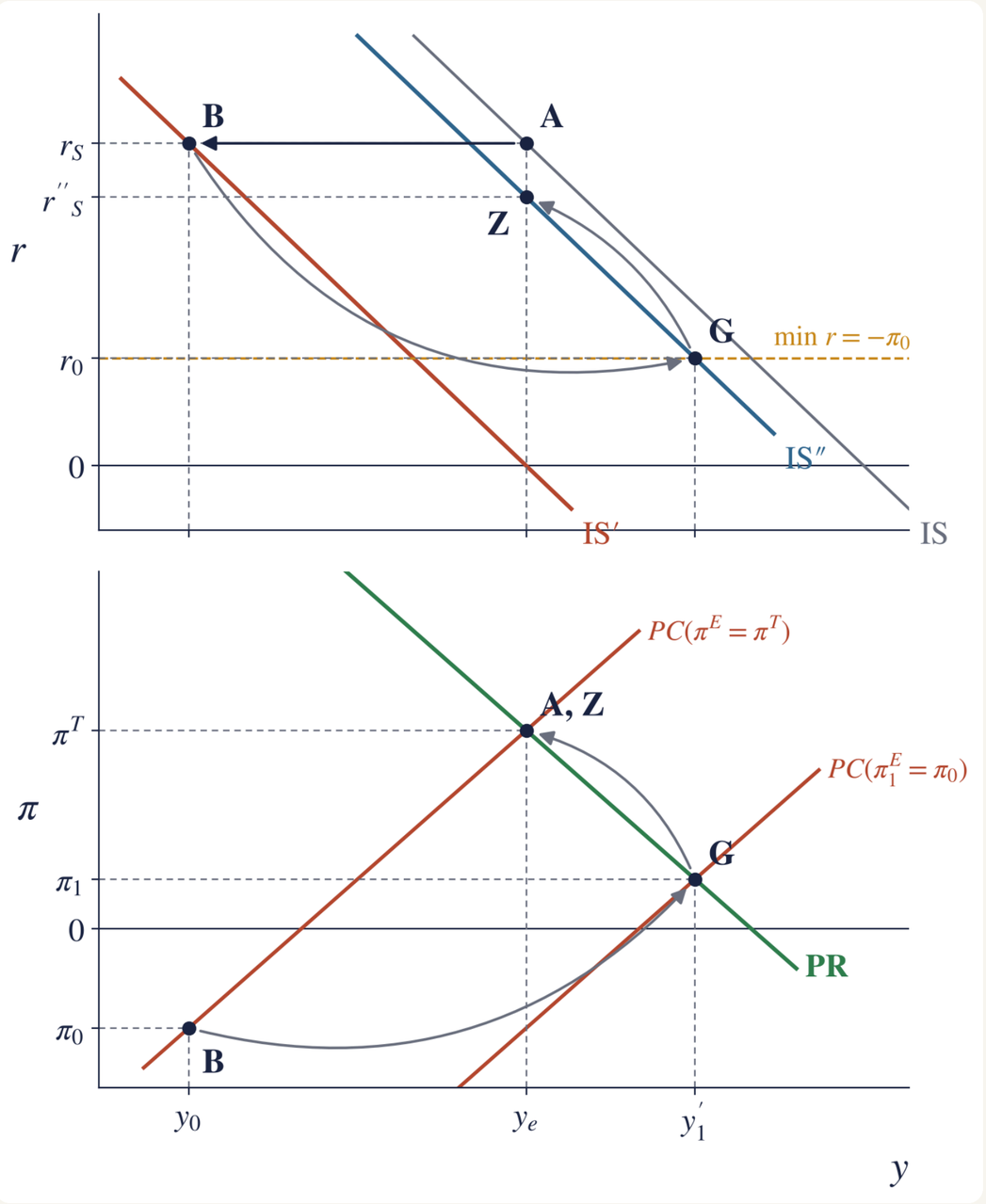
Period 1 · B → C

Output only reaches y_1, far below the target y'_1. Still a negative gap, so inflation falls further, to π_1. The floor rises to $-\pi_1$.

Period 2 · C → D and beyond

The real rate is forced **up**, output falls, deflation deepens. No return to A.

$\pi \downarrow \rightarrow \min r \uparrow \rightarrow y \downarrow \rightarrow \pi \downarrow\downarrow$



QUESTION 6(B) · ESCAPING THE TRAP

Fiscal policy moves the IS for the bank

Raise *G* so the IS shifts to **IS″**, through G. Now the lowest rate it can set, $r_0 = -\pi_0$, delivers the target output y'_1.

Monetary and fiscal policy now work together, so MR becomes a policy rule, **PR**. The positive gap lifts inflation; the bank raises *r* step by step to Z.

At Z: y_e and π^T again, with higher *G* and a stabilising rate r''_s below the original.

Other routes, and their limits

Raise expected inflation, so the floor $-\pi^E$ falls. Quantitative easing (Chapters 5–6). Limits: spending may not respond to lower rates, and fiscal space depends on the public finances.

QUESTION 6(C) · THE UK AFTER 2008

Two likely policy mistakes

1

Leaning on monetary policy at the floor

Bank Rate was cut to 0.5% in March 2009. With little room left to cut, real rates could not fall far enough, and output stayed below equilibrium.

2

Tightening fiscal policy too early

Austerity from 2010 shifted the IS further **left**, the opposite of the move to IS″ that the escape route needs.

For discussion: the model also misses why the UK settled *below* the old path. If a long slump lowers y_e itself (hysteresis, Chapter 4), the economy “recovers” to a worse equilibrium.

Questions?

Email me any time: **MUB@CARDIFF.AC.UK**

Next week · Lecture 4: expectations and monetary policy